

Input and Output

Where We Left Off

We can store, compute, repeat, and organize

But every result has stayed **inside** the program

So far the only way to see it has been the **debugger**

Today: talking to the outside world

What I/O Is

Input: data coming **into** the program
keyboard, files, sensors, network

Output: data going **out** of the program
screen, files, actuators, network

I/O is how a program serves its **users**

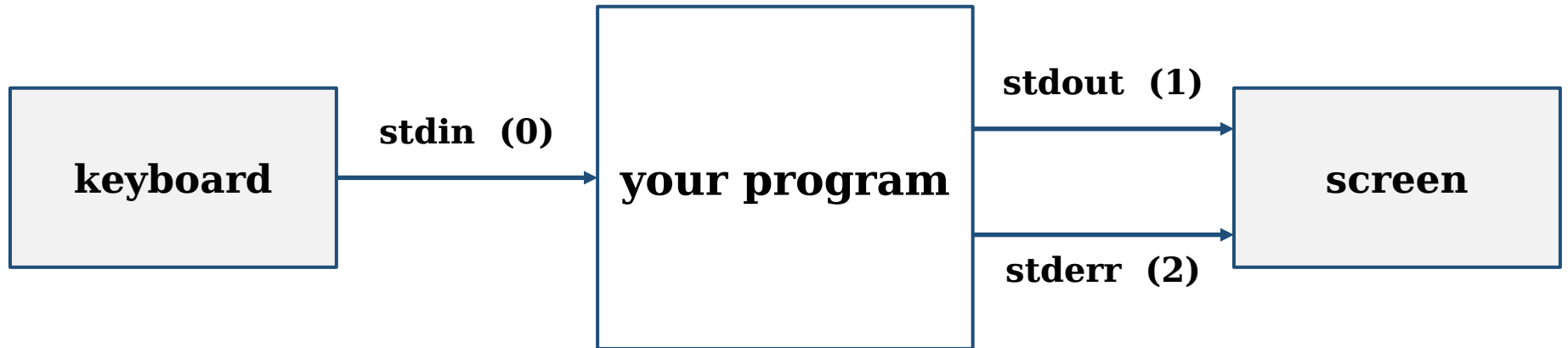
Streams

A **stream** is a sequence of bytes flowing into or out of a program

Every program starts with **three**:

stdin	0	standard input
stdout	1	standard output
stderr	2	standard error

The Three Standard Streams



By default all three are connected to the terminal

Why a Separate Error Stream?

`./prog > results.txt`
stdout goes to the file
stderr **still** appears on the screen

`./prog > results.txt 2> errors.txt`
each stream to its own file

Data and error messages never mix

Output in C: printf

```
#include <stdio.h>
```

```
printf ( "Oil pressure\n" ) ;
```

```
printf ( "psi = %f\n", psi ) ;
```

```
printf ( "%u counts = %.1f psi\n", counts, psi ) ;
```

% marks where a value is inserted

\n ends the line

Format Specifiers

%d	signed int	%x	hexadecimal
%u	unsigned int	%c	one character
%ld	long	%s	string
%f	float or double	%p	pointer address
%e	scientific	%%	a literal %

The specifier must **match** the type

Width and Precision

```
printf ( "[%5d]", 42 ) ;      [  42]
printf ( "[%5d]", 42 ) ;      [42  ]
printf ( "[%05d]", 42 ) ;     [00042]
printf ( "[%1f]", 50.0 ) ;     [50.0]
printf ( "[%8.2f]", 3.14159 ) ; [  3.14]
printf ( "[%#x]", 255 ) ;      [0xff]
```

Width pads, precision sets digits after the point

Printing Fixed-Width Types

uint32_t may not be unsigned int on every machine

```
#include <inttypes.h>
```

```
printf ( "%u" PRIu32 "\n", counts ) ;  
printf ( "%u" PRIu64 "\n", timestamp_ms ) ;
```

PRIu32 expands to the right specifier

A Mismatch Is a Bug

```
printf ( "%d\n", 3.5 ) ;    prints garbage  
printf ( "%s\n", 42 ) ;    may crash
```

printf trusts the format string completely

Compile with **-Wall** and the compiler
checks every format for you

More C Output Functions

`fprintf ( stderr, "sensor fault\n" ) ;` to stderr
`puts ( "done" ) ;` string + newline
`putchar ( 'x' ) ;` one character
`sprintf ( buf, "%.1f", psi ) ;` into a string
`snprintf ( buf, sizeof buf, "%.1f", psi ) ;`

Prefer **snprintf** - it cannot overflow buf

Output in C++: iostream

```
#include <iostream>
```

```
std::cout << "Oil pressure\n" ;
```

```
std::cout << "psi = " << psi << '\n' ;
```

```
std::cerr << "sensor fault\n" ;
```

<< sends each value to the stream in turn

Formatting in C++

```
#include <iomanip>
```

```
std::cout << std::fixed << std::setprecision ( 1 )  
          << psi << '\n' ;                50.0
```

```
std::cout << std::setw ( 8 ) << counts ;    1456  
std::cout << std::hex << 255 ;             ff
```

Careful: most settings **stay on** until changed

Modern C++: `std::format`

C++20:

```
#include <format>
std::cout << std::format ( "{:.1f} psi\n", psi ) ;
```

C++23:

```
#include <print>
std::println ( "{:.1f} psi", psi ) ;
```

Checked at **compile time**, like Rust

Output in Rust: println!

```
println! ( "Oil pressure" ) ;  
println! ( "psi = {}", psi ) ;  
println! ( "{} counts = {:.1} psi", counts, psi ) ;
```

print!	no newline
eprintln!	to stderr

The **!** means a **macro**, checked at compile time

Rust Format Strings

<code>{:.1}</code>	50.0	one decimal place
<code>{:8.2}</code>	3.14	width 8
<code>{:<8}</code>	42	left align
<code>{:08}</code>	00000042	pad with zeros
<code>{:x}</code> <code>{:#x}</code>	ff 0xff	hexadecimal
<code>{:b}</code>	11111111	binary

Name the variable inline: `println! ( "{psi:.1}" ) ;`

Printing a Struct in Rust

```
#[derive(Debug)]  
struct Sample { adc_counts : u32, psi : f32 }  
  
println! ( "{:?}", my_sample ) ;  
    Sample { adc_counts: 1456, psi: 50.0 }  
  
{:#?} prints one field per line
```

Printing a Struct in C and C++

C: write a function

```
void print_sample ( const sample_t *s ) {  
    printf ( "%u counts, %.1f psi\n",  
            s->adc_counts, s->psi ) ;  
}
```

C++: overload operator<< so that
std::cout << my_sample ; works

One Line, Three Languages

C: `printf ( "%.1f psi\n", psi ) ;`

C++: `std::cout << std::fixed << std::setprecision
(1)
 << psi << " psi\n" ;`

Rust: `println! ( "{:.1} psi", psi ) ;`

Output Is Buffered

Output is collected in a **buffer** and sent in chunks

stdout to a terminal:

sent at each newline

stdout to a file:

sent when the buffer fills

stderr:

sent immediately

Force it out:

```
fflush ( stdout ) ;
```

```
std::cout << std::flush ;
```

Buffering Can Lie to You

```
printf ( "got here" ) ;      no newline  
crash ( ) ;
```

The program crashes - and "got here"
is **never printed**

It was still waiting in the buffer
You conclude the crash happened **earlier**

Print Is Not a Debugger

Output is for the program's **users**

It is **not** a substitute for the debugger

Why not?

The Temptation

```
printf ( "here 1\n" ) ;  
x = compute ( a, b ) ;  
printf ( "here 2, x = %d\n", x ) ;  
y = scale ( x ) ;  
printf ( "here 3, y = %d\n", y ) ;
```

It feels quick. It is not.

What Goes Wrong

Every new question means **edit, compile, run**

You only see what you **guessed** to print

Buffered output can vanish in a crash

Forgotten prints end up in the final program

Printing **changes** the program you are testing

Heisenbugs

Add a print statement - and the bug **disappears**

The print changed:

- the timing of the program

- the layout of the stack

- what the optimizer did

Remove the print - and the bug is back

On a Microcontroller

Printing goes out a **serial port**

115200 baud = about 11,500 characters per second

One 40-character line = about **3.5 ms**

Enough to break real-time timing

Never print inside an interrupt routine

Ask the Debugger Instead

The question	The debugger answer
Did the program get here?	<code>break file:line</code>
What is x right now?	<code>print x</code>
What are all the locals?	<code>info locals</code>
How did I get here?	<code>backtrace</code>
Who changed x?	<code>watch x</code>
Only on pass 97	<code>break ... if i == 97</code>
Show sum at every stop	<code>display sum</code>

The Same Question, Two Ways

With print: `add printf ( "i=%d sum=%d\n", i, sum ) ;`
 `recompile, rerun, read 100 lines`

With gdb: `break main.c:12 if i == 97`
 `display sum`
 `run`

No edit. No recompile. One stop, exactly where you asked.

Our Rule

Print statements are for the program's **users**

Debugging is done with the **debugger**

Any print added while debugging is **removed** before you submit

When Output Is the Right Tool

Logging in a system that runs for days
or in the field, where no debugger is attached

Planned, not sprinkled:

- a log file or stderr

- severity levels: error, warning, info

- can be switched off

A design decision - not a debugging shortcut

Input: Reading From the Keyboard

The program **waits** until the user types a line and presses Enter

What arrives is **text** - characters

It must be **converted** to a number

And the user may type anything

Input in C: scanf

```
int count ;  
scanf ( "%d", &count ) ;
```

scanf needs the **address** of count
so it can store into (count)

```
float:      scanf ( "%f", &psi ) ;  
double:    scanf ( "%lf", &value ) ; not %f !
```

scanf Pitfalls

Returns how many values it converted:

```
if ( scanf ( "%d", &count ) != 1 ) { /* bad input
*/ }
```

Bad input stays in the buffer - the next
scanf fails on it too

Strings: always give a width

```
char name[20] ; scanf ( "%19s", name ) ;
```

Safer Input in C

Read a whole line, then convert it:

```
char line[64] ;  
if ( fgets ( line, sizeof line, stdin ) != NULL ) {  
    char *end ;  
    long v = strtol ( line, &end, 10 ) ;  
    if ( end == line ) { /* not a number */ }  
}
```

Input in C++: cin

```
int count ;  
if ( std::cin >> count ) {  
    // got a number  
} else {  
    // not a number - the stream is now in a fail state  
}
```

Whole line: `std::getline ( std::cin, line ) ;`

Input in Rust

```
let mut line = String::new ( ) ;  
std::io::stdin ( ).read_line ( &mut line )  
    .expect ( "read failed" ) ;
```

```
let count : i32 = line.trim ( ).parse ( )  
    .expect ( "not a number" ) ;
```

&mut line - read_line may change it

Handling Bad Input in Rust

```
match line.trim ( ).parse::<i32> ( ) {  
    Ok ( n )    => println! ( "got {}", n ),  
    Err ( _ )   => eprintln! ( "not a number" ),  
}
```

parse returns a **Result**: Ok with a value,
or Err with the reason

The compiler makes you handle **both**

Input Can Always Fail

The user types letters instead of digits

The input ends (Ctrl-D, end of file)

The file does not exist

The sensor is unplugged

Always check:

C return value

C++ stream state

Rust Result

Reading Until the End

C:

```
while ( fgets ( line, sizeof line, stdin ) != NULL )  
{ ... }
```

C++:

```
while ( std::getline ( std::cin, line ) ) { ... }
```

Rust:

```
for line in std::io::stdin ( ).lines ( ) {  
    let line = line.expect ( "read failed" ) ;  
}
```

Files Are Streams Too

Open a file to get a stream

Read or **write** it like stdin and stdout

Close it when you are done

Everything you learned about printing
applies

Writing a File in C

```
FILE *fp = fopen ( "log.csv", "w" ) ;  
if ( fp == NULL ) {  
    perror ( "log.csv" ) ;  
    return 1 ;  
}  
fprintf ( fp, "%u,%.1f\n", counts, psi ) ;  
fclose ( fp ) ;
```

fopen Modes

"r" read - the file must exist
"w" write - create, or **erase** an existing file
"a" append - add to the end
"r+" read and write

Add **b** for binary: "rb" "wb"

Reading a File in C

```
FILE *fp = fopen ( "log.csv", "r" ) ;  
if ( fp == NULL ) { perror ( "log.csv" ) ; return 1 ; }  
  
char line[128] ;  
while ( fgets ( line, sizeof line, fp ) != NULL ) {  
    unsigned counts ; float psi ;  
    if ( sscanf ( line, "%u,%f", &counts, &psi ) == 2 )  
    { ... }  
}  
fclose ( fp ) ;
```

Files in C++

```
#include <fstream>
```

```
std::ofstream out ( "log.csv" );  
if ( !out ) { std::cerr << "cannot open\n" ; return  
1 ; }
```

```
out << counts << ',' << psi << '\n' ;
```

Closed **automatically** at the closing brace

Files in Rust

```
use std::fs::File ;  
use std::io::Write ;
```

```
let mut f = File::create ( "log.csv" )? ;  
writeln! ( f, "{} {:.1}", counts, psi )? ;
```

? passes any error back to the caller
Closed **automatically** when `f` goes out of scope

Always Close Your Files

C: forget `fclose` and buffered data
may

never reach the disk

C++, Rust: the file closes when its owner
goes out of **scope**

The operating system limits how many
files can be open at once

Example: Log Samples in C

```
FILE *fp = fopen ( "log.csv", "w" ) ;  
if ( fp == NULL ) { perror ( "log.csv" ) ; return 1 ; }  
  
fprintf ( fp, "adc_counts,psi\n" ) ;  
for ( int i = 0 ; i < n ; i++ ) {  
    fprintf ( fp, "%" PRIu32 ",%.1f\n",  
             history[i].adc_counts, history[i].psi ) ;  
}  
fclose ( fp ) ;
```

Example: Log Samples in Rust

```
fn save ( history : &[Sample] ) -> std::io::Result<()> {  
    let mut f = File::create ( "log.csv" )? ;  
    writeln! ( f, "adc_counts,psi" )? ;  
    for s in history.iter ( ) {  
        writeln! ( f, "{} {:.1}", s.adc_counts, s.psi )? ;  
    }  
    Ok ( ( ) )  
}
```

Text or Binary?

Text: `fprintf ( fp, "%u,%.1f\n", ... ) ;`
human readable, portable, larger

Binary: `fwrite ( &s, sizeof s, 1, fp ) ;`
the raw bytes of the struct
includes **padding**, depends on byte order

Prefer text unless size or speed demands binary

I/O on a Microcontroller

No screen and no keyboard

I/O means **pins**, the **ADC**, and a **serial port**

`Serial.println ( psi ) ;` Arduino output

Many boards, including the ESP32-S3, have a **hardware debugger** built in over USB

The Same Rule on Hardware

Serial output is for **monitoring** the device

Debugging uses the **hardware debugger**:
breakpoints on the real chip
watch registers and variables
no change to the program's timing

Print statements are still not a debugger

Output Compared

	C	C++	Rust
to stdout	printf	std::cout <<	println!
to stderr	fprintf (stderr, ...)	std::cerr <<	eprintln!
format	%d %.1f %x	iomanip / std::format	{ } {:.1} {:x}
to a file	fprintf (fp, ...)	std::ofstream	writeln! (f, ...)
to a string	snprintf	std::format	format!
flush	fflush	std::flush	.flush ()

Input Compared

	C	C++	Rust
a number	<code>scanf ("%d", &n)</code>	<code>std::cin >> n</code>	<code>read_line + parse</code>
a line	<code>fgets</code>	<code>std::getline</code>	<code>read_line</code>
convert	<code>strtol, sscanf</code>	<code>std::stoi</code>	<code>.parse ()</code>
open a file	<code>fopen</code>	<code>std::ifstream</code>	<code>File::open</code>
check it	return value	stream state	Result
close	<code>fclose</code>	end of scope	end of scope

What We Can Now Do

Print formatted results in all three languages

Read and check user input

Read and **write** files

And keep print statements for users -
the **debugger** is for debugging

Let's practice

Exercise: Using **iopractice**